

SIMATS ENGINEERING

COURSE NAME: TRANSMISSION LINES AND WAVE GUIDES

COURSE CODE: ECA1505

COURSE OUTCOME COVERED: CO3AT1

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Using MATLAB/Scilab, simulate movement along a $50\ \Omega$ transmission line for a load impedance of $100 - j50\ \Omega$. Plot the impedance trajectory on the Smith chart for distances from 0 to $\lambda/2$.

Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Parameters
```

```
Z0 = 50;           % Characteristic impedance (ohms)
```

```
ZL = 100 - 1j*50;  % Load impedance (ohms)
```

```
%% Normalize load impedance
```

```
zL = ZL/Z0;
```

```
%% Load reflection coefficient
```

```
GammaL = (zL - 1)/(zL + 1);
```

```
%% Distance from 0 to  $\lambda/2$ 
```

```
d = linspace(0, 0.5, 1000); % d/lambda
```

```

%% Reflection coefficient along transmission line
Gamma = GammaL .* exp(-1j*4*pi*d);

%% Impedance trajectory
z = (1 + Gamma)./(1 - Gamma);

%% Convert to actual impedance
Z = Z0*z;

%% Display important values
fprintf('Normalized Load Impedance = %.2f %+.2fj\n', ...
    real(zL), imag(zL));

fprintf('Load Reflection Coefficient = %.4f %+.4fj\n', ...
    real(GammaL), imag(GammaL));

fprintf('|Gamma| = %.4f\n', abs(GammaL));

VSWR = (1 + abs(GammaL))/(1 - abs(GammaL));
fprintf('VSWR = %.4f\n', VSWR);

%% Plot Smith chart
figure;

% Draw Smith chart background
theta = linspace(0,2*pi,1000);
plot(cos(theta), sin(theta), 'k', 'LineWidth', 1.5);
hold on;

```

```

grid on;

axis equal;

% Plot constant reflection coefficient circle
plot(real(Gamma), imag(Gamma), 'b', 'LineWidth', 2);

% Mark load point
plot(real(GammaL), imag(GammaL), 'ro', ...
      'MarkerSize', 8, 'LineWidth', 2);

% Mark halfway point
mid = 501;
plot(real(Gamma(mid)), imag(Gamma(mid)), 'gs', ...
      'MarkerSize', 8, 'LineWidth', 2);

xlabel('Real(\Gamma)');
ylabel('Imaginary(\Gamma)');
title('Impedance Trajectory for  $0 \leq d \leq \lambda/2$ ');

legend('Smith Chart Boundary', ...
      'Impedance Trajectory', ...
      'Load Point', ...
      'd =  $\lambda/4$ ');

xlim([-1.1 1.1]);
ylim([-1.1 1.1]);

```

OUTPUT:

